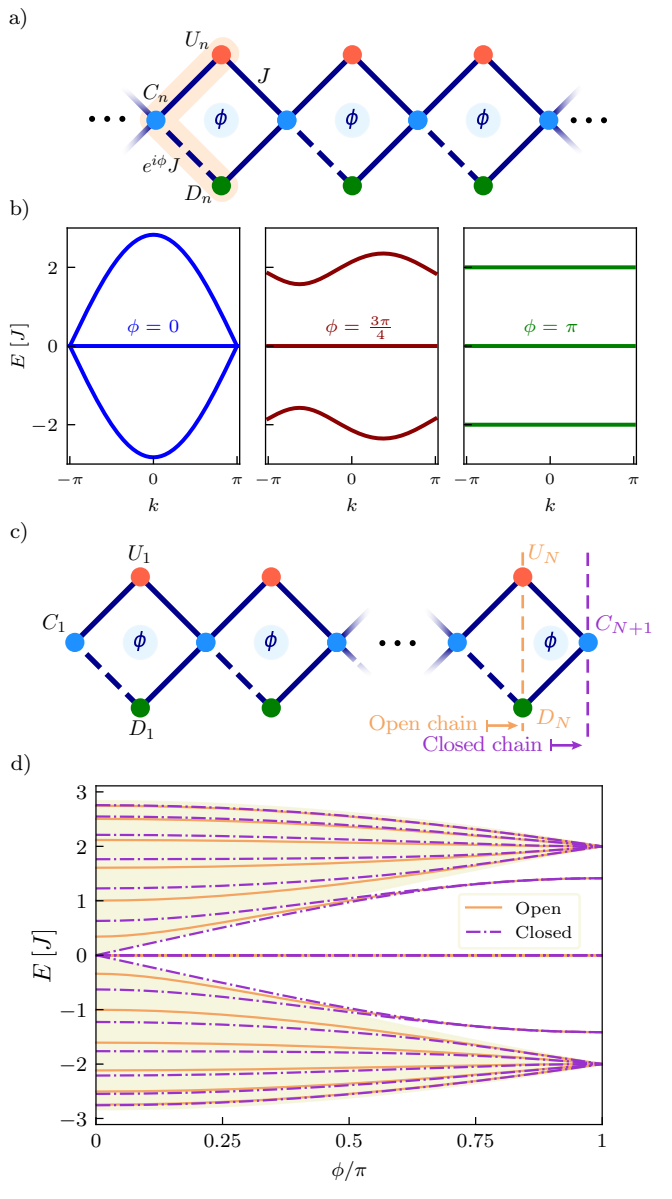

a)
$U_n$
$J$
$C_n$
$\phi$
$\phi$
$\phi$
$e^{i\phi}J$
$D_n$
b)
$E\,[J]$
2
0
−2
$\phi = 0$
$\phi = \frac{3\pi}{4}$
$\phi = \pi$
$-\pi$
0
$\pi$
$k$
c)
$U_1$
$C_1$
$\phi$
$\phi$
$D_1$
$U_N$
$C_{N+1}$
$\phi$
Open chain
$D_N$
Closed chain
d)
$E\,[J]$
3
2
1
0
−1
−2
−3
Open
Closed
0
0.25
0.5
0.75
1
$\phi/\pi$